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PROJECT REPORT ON

**“ArogyaSathi: Offline Mobile EHR
for ASHA Workers”**

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ArogyaSathi: Offline Mobile EHR for ASHA Workers

Abstract

Rural healthcare workers, particularly Accredited Social Health Activists (ASHA), play a crucial role in collecting and maintaining patient health data within village communities. However, many rural regions face unreliable or non-existent internet connectivity, forcing healthcare workers to rely heavily on paper-based record systems. These manual methods are prone to errors, data loss, duplication, and delays in reporting, which negatively affect patient monitoring and timely healthcare interventions. This project addresses these challenges by developing ArogyaSathi, an offline-first mobile Electronic Health Record (EHR) system specifically designed for ASHA workers operating in low-connectivity environments.

The primary objective of this project is to provide a lightweight and reliable digital solution that enables healthcare workers to register patients, record health visits, track vaccinations, and manage antenatal care (ANC) information directly on a mobile device without requiring continuous internet connectivity. The system is developed using Flutter for mobile application development, SQLite for local offline data storage, and FastAPI as the backend framework for server-side services. A synchronization mechanism ensures that locally stored data can be securely transmitted to a centralized server when internet connectivity becomes available.

The developed prototype demonstrates that an offline-first architecture can effectively support healthcare data collection and management in rural settings while reducing reliance on paper documentation. By enabling reliable record management and automated reminders for healthcare follow-ups, ArogyaSathi improves the efficiency of field healthcare workers and enhances the accessibility of patient data for healthcare administrators. The project highlights the potential of mobile health technologies to strengthen rural healthcare systems and contribute to scalable digital health solutions for underserved communities.

1. INTRODUCTION

Healthcare delivery in rural and remote regions of India heavily relies on Accredited Social Health Activists (ASHA) and Primary Health Centre (PHC) staff. These frontline healthcare workers are responsible for maternal and child health monitoring, vaccination follow-ups, and maintaining essential health records for community members. Their role is critical in ensuring the effective implementation of public health programs at the grassroots level.

Despite their importance, healthcare workers in rural areas face several operational challenges, particularly due to poor and inconsistent internet connectivity. Most existing Electronic Health Record (EHR) systems depend on continuous internet access, making them unsuitable for use in remote environments where network infrastructure is limited or unreliable.

As a result, healthcare data is frequently recorded manually using paper-based registers. This manual process often leads to problems such as data duplication, loss of records, delayed reporting, and difficulty in maintaining patient history. These issues reduce the efficiency of healthcare workers and make it difficult for healthcare administrators to access accurate and timely data.

To address these challenges, this project proposes ArogyaSathi, an offline-first mobile Electronic Health Record companion application designed to assist ASHA workers in collecting, storing, and managing patient health records efficiently, even in environments with limited or no internet connectivity.

1.1. Motivation

Rural healthcare systems rely heavily on ASHA workers to collect and maintain essential patient health records, including vaccination details, antenatal care information, and follow-up visits. In many rural areas, these records are still maintained using paper-based registers due to unreliable or limited internet connectivity. Paper records are difficult to manage, prone to damage or loss, and often result in delayed reporting and inefficient patient tracking. Additionally, most existing digital health record systems require continuous internet access, making them unsuitable for use in remote environments. These challenges highlight the need for a reliable digital solution that can function effectively without constant connectivity. The motivation behind developing **ArogyaSathi** is to provide ASHA workers with an offline-first mobile application that enables efficient patient data collection, local record management, and automatic synchronization with centralized healthcare systems when internet connectivity becomes available.

1.2. Problem Statement and Proposed Solution

1.2.1. **Problem Statement:** Most existing digital healthcare record systems require continuous internet connectivity, which is not feasible in many rural and remote areas. ASHA workers face difficulties in maintaining accurate electronic health records during field visits due to network limitations. This leads to inefficient data management, delayed healthcare interventions, and increased administrative workload.

1.2.2. **Proposed Solution:** The proposed solution, *ArogyaSathi*, is a lightweight, offline-first mobile Electronic Health Record (EHR) application designed to support ASHA workers in rural and low-connectivity environments.

- ❖ To develop an offline-first mobile EHR application tailored for ASHA workers.
- ❖ To enable secure local storage of patient health records on the mobile device.
- ❖ To implement automatic data synchronization when internet connectivity is restored.
- ❖ To provide multilingual user interface support for improved accessibility.
- ❖ To generate vaccination and ANC follow-up reminders to ensure timely healthcare delivery.
- ❖ To offer role-based dashboards for ASHA workers and PHC staff.
- ❖ To ensure a simple, intuitive, and user-friendly interface requiring minimal training.

1.3. Purpose, Objectives, Goals

1.3.1. **Purpose:** The purpose of this project is to develop a reliable and user-friendly mobile EHR system that improves healthcare data collection and continuity of care in low-connectivity rural environments.

1.3.2. Objectives

- ❖ To design and develop an offline-first mobile EHR application
- ❖ To enable secure local storage of patient health records
- ❖ To implement automatic data synchronization when internet connectivity is restored
- ❖ To provide multilingual support for better accessibility
- ❖ To generate reminders for vaccination and ANC follow-ups
- ❖ To provide role-based access for ASHA workers and PHC staff

1.3.3. Goals

- ❖ To improve accuracy and availability of rural healthcare data
- ❖ To reduce dependency on paper-based records
- ❖ To enhance follow-up care and patient tracking
- ❖ To provide a scalable and cost-effective digital health solution

1.4. Literature Survey

The use of digital technologies in healthcare has significantly improved the management, accessibility, and accuracy of patient health information. Electronic Health Record (EHR) systems are widely adopted in hospitals and healthcare institutions to store and manage patient medical data electronically. These systems enable efficient data sharing among healthcare professionals and support better clinical decision-making. However, many traditional EHR systems are designed for environments with stable internet connectivity and advanced technological infrastructure, which limits their applicability in rural healthcare settings [1].

To improve healthcare accessibility in remote regions, researchers have explored mobile health (mHealth) technologies, which utilize mobile devices to support healthcare services and data collection. Studies have shown that mHealth applications can enhance healthcare delivery by enabling field health workers to collect patient information, monitor vaccination schedules, and report health indicators using mobile devices [2]. Mobile health solutions have been widely used in maternal and child health programs, particularly in developing countries, where they help improve reporting accuracy and communication between community health workers and healthcare institutions.

Despite these advantages, several studies indicate that many existing mHealth systems rely heavily on continuous internet connectivity for real-time data transmission and storage [3]. In rural areas where network connectivity is unreliable or unavailable, this dependence creates significant operational challenges for healthcare workers. As a result, health workers often revert to manual record-keeping methods, which reduces efficiency and increases the risk of data loss or duplication.

Recent research emphasizes the importance of offline-first application architectures for healthcare systems operating in low-connectivity environments. Offline-first systems allow healthcare workers to store patient data locally on mobile devices and synchronize the information with centralized databases when internet connectivity becomes available. This approach ensures uninterrupted data collection while maintaining data consistency across healthcare systems [4].

Although these approaches have improved digital healthcare delivery, there remains a need for simple and user-friendly solutions tailored specifically for community health workers such as ASHA workers in rural India. The ArogyaSathi system addresses this gap by providing an offline-first mobile Electronic Health Record application that enables efficient patient data management, visit recording, vaccination tracking, and reminder generation while ensuring reliable data synchronization with centralized healthcare systems.

1.5. Project Scope and Limitations

1.5.1. **Scope:** The scope of the project *ArogyaSathi* is limited to developing an offline-first mobile EHR application for ASHA workers to record and manage basic healthcare data in low-connectivity rural areas.

- ❖ **Purpose:** The project focuses on developing an offline-first mobile Electronic Health Record (EHR) application to assist ASHA workers in recording and managing essential healthcare data in rural and low-connectivity regions.
- ❖ **Functionality:** The system supports offline data entry and local storage of patient health records, ensuring uninterrupted usage during field visits.
- ❖ **Data Synchronization:** The application enables automatic synchronization of locally stored data with a central system when internet connectivity becomes available.
- ❖ **Accessibility:** Multilingual user interface support is provided to enhance usability and adoption among ASHA workers.
- ❖ **Healthcare Support:** The system includes reminder notifications for vaccinations and antenatal care (ANC) follow-ups to support timely healthcare delivery.
- ❖ **User Roles:** Role-based access is incorporated to allow ASHA workers and PHC staff to view and manage relevant healthcare information.

1.5.2. Limitations

The proposed system is developed as a functional prototype and therefore has certain limitations. It does not aim to replace full-scale hospital EHR systems and requires internet connectivity for data synchronization. Integration with external government health databases and advanced clinical features are outside the scope of this project.

- ❖ **Prototype Scope:** The system is developed as a functional prototype and is not intended to replace full-scale, certified hospital EHR systems.
- ❖ **Connectivity Requirement:** Internet connectivity is required only for synchronizing data with the backend system.
- ❖ **System Integration:** Integration with national or state-level health information systems is not included in the current scope.

- ❖ **Advanced Capabilities:** Features such as clinical decision support, diagnostics, and advanced data analytics are beyond the scope of this project. information.

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2. System Analysis

Healthcare information management systems are used to record, store, and manage patient health data to support healthcare service delivery. In urban healthcare environments, Electronic Health Record (EHR) systems and digital health platforms are widely used to maintain centralized patient records and facilitate data sharing among healthcare providers.

In rural healthcare settings, community health workers such as ASHA workers collect and maintain essential health data related to patient registration, vaccination tracking, maternal health monitoring, and follow-up visits. While digital healthcare systems are increasingly being introduced, their effectiveness in rural environments is often limited by infrastructure constraints and operational challenges.

To understand the requirements of the proposed system, it is important to analyze the existing methods used for healthcare data management, identify their limitations, and determine the functional needs of healthcare workers and administrators. This analysis provides the basis for defining the system requirements and guiding the design of the proposed **ArogyaSathi** system.

2.1. Existing Systems

Healthcare data management for maternal and community health programs in rural India is currently supported through the following systems:

2.1.1. Paper-Based Health Registers

ASHA workers maintain physical registers to record patient information, maternal and child health details, immunization schedules, and follow-up visits during field activities.

2.1.2. Health Management Information System (HMIS)

A web-based reporting platform under the National Health Mission used by healthcare institutions to submit aggregated health data related to service delivery.

2.1.3. Reproductive and Child Health (RCH) Portal

A centralized system used by healthcare facilities to track maternal and child health services, including pregnancy registration, antenatal care, and immunization records.

2.2. Scope and Limitations of Existing Systems

While these systems support healthcare data recording and reporting, they present several limitations in rural field environments:

2.2.1. Dependence on Manual Records

ASHA workers primarily rely on paper registers for field data collection, which increases the risk of record loss, duplication, and manual errors.

2.2.2. **Limited Field-Level Digital Access**

Government digital platforms such as HMIS and the RCH portal are mainly used at healthcare facilities and are not designed for direct use during field visits.

2.2.3. **Internet Connectivity Requirement**

Digital reporting systems typically require internet connectivity, which may be unreliable in rural areas.

2.2.4. **Delayed Data Reporting**

Field data collected in paper registers must later be manually transferred into digital systems at healthcare facilities, causing delays in reporting and data availability.

These limitations highlight the need for a system that allows **direct digital data collection in the field while functioning in low-connectivity environments**, which is addressed by the proposed **ArogyaSathi** system.

2.3. **Project Perspective and Features**

The proposed ArogyaSathi system is designed as a mobile application to support ASHA workers in collecting and managing healthcare data during field visits. The system focuses on enabling digital health record management in rural environments where internet connectivity may be limited. By allowing healthcare workers to store patient information locally on the device and synchronize it with a central system when connectivity becomes available, the application improves the efficiency and reliability of healthcare data collection.

2.3.1. **Features**

- ❖ **Patient Registration:** – Recording and managing basic patient information.
- ❖ **Health Visit Records:** – Maintaining records of patient visits, health observations, and vaccination status.
- ❖ **Offline Data Storage:** – Storing healthcare data locally on the mobile device to allow operation without internet connectivity.
- ❖ **Data Synchronization:** – Synchronizing locally stored data with a backend system when internet connectivity is available.
- ❖ **Reminder Support:** – Providing reminders for vaccination schedules and follow-up visits.
- ❖ **Multilingual Interface:** – Supporting English and Marathi language options for ease of use.
- ❖ **User-Friendly Interface:** – Clean and intuitive design for efficient use by ASHA workers during field activities.

2.4. Stakeholders

The development and use of the ArogyaSathi system involve the following stakeholders:

- ❖ ASHA Workers – Primary users who record patient information, health visit details, and vaccination data during field visits.
- ❖ Primary Health Centre (PHC) Staff – Healthcare personnel who review and monitor health records collected from field workers.
- ❖ Healthcare Administrators – Authorities responsible for monitoring healthcare programs and analysing community health data.
- ❖ System Developers – Responsible for developing, maintaining, and improving the application.

2.5. Requirement Analysis

Requirement analysis identifies the essential system capabilities needed to support healthcare data collection and management in rural environments.

2.5.1. Functional Requirements

- ❖ Register and manage patient information.
- ❖ Record patient visit details and health observations.
- ❖ Maintain vaccination and follow-up records.
- ❖ Store healthcare data locally on the mobile device.
- ❖ Synchronize stored data with a central system when internet connectivity is available.
- ❖ Provide reminder support for scheduled healthcare activities.
- ❖ Support multilingual interface options (English and Marathi).

2.5.2. Performance Requirements

- ❖ The application should operate efficiently on standard Android mobile devices.
- ❖ Data entry and retrieval should be fast to support field operations.
- ❖ The system should function reliably without continuous internet connectivity.

2.5.3. Security Requirements

- ❖ Access to the application should be restricted to authorized users.
- ❖ Patient information should be stored securely to prevent unauthorized access.
- ❖ Data synchronization with the backend system should use secure communication methods.

3. System Design

- ❖ To design an offline-first system architecture that allows data collection without continuous internet access.
- ❖ To define a client-server model where the mobile application acts as the primary data collection unit.
- ❖ To ensure secure handling of healthcare data during both offline storage and synchronization.

3.1. Design Constraints

The design of the **ArogyaSathi** system is subject to the following constraints:

- ❖ The system must be implemented as a mobile application for Android devices, as ASHA workers primarily use smartphones during field activities.
- ❖ The system must operate in low-connectivity environments, requiring support for offline data entry and delayed data synchronization.
- ❖ The application must be developed using the selected technology stack, including Flutter for the mobile application and SQLite for local data storage.
- ❖ The system must provide a simple and intuitive user interface to support efficient use by field healthcare workers.
- ❖ The application must support multilingual interaction, specifically English and Marathi, to improve usability.

3.2. System Architecture

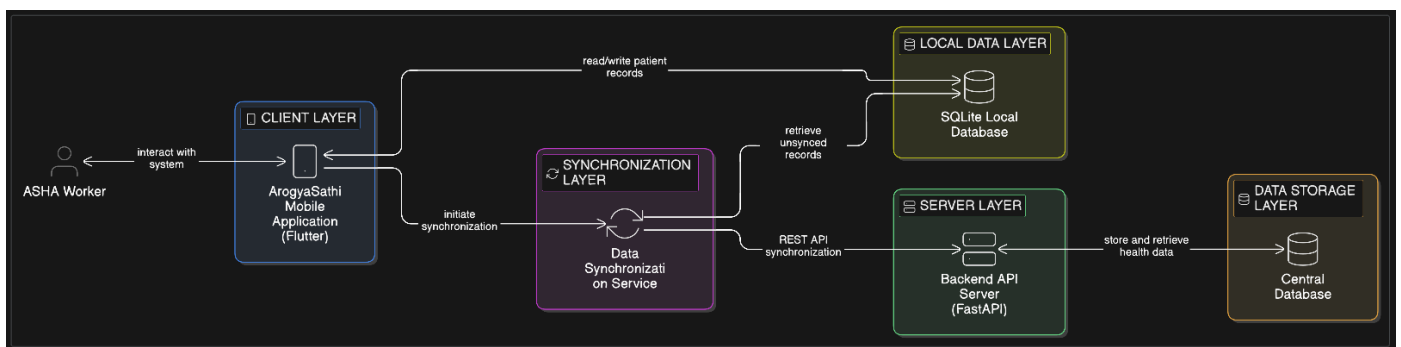


Figure 1: System Architecture of the ArogyaSathi Offline-First Mobile Health System

3.3. ER Diagram



Figure 2:- ER Diagram of the ArogyaSathi Offline-First Mobile Health System

3.4. User Interface (UI) Navigation Flow

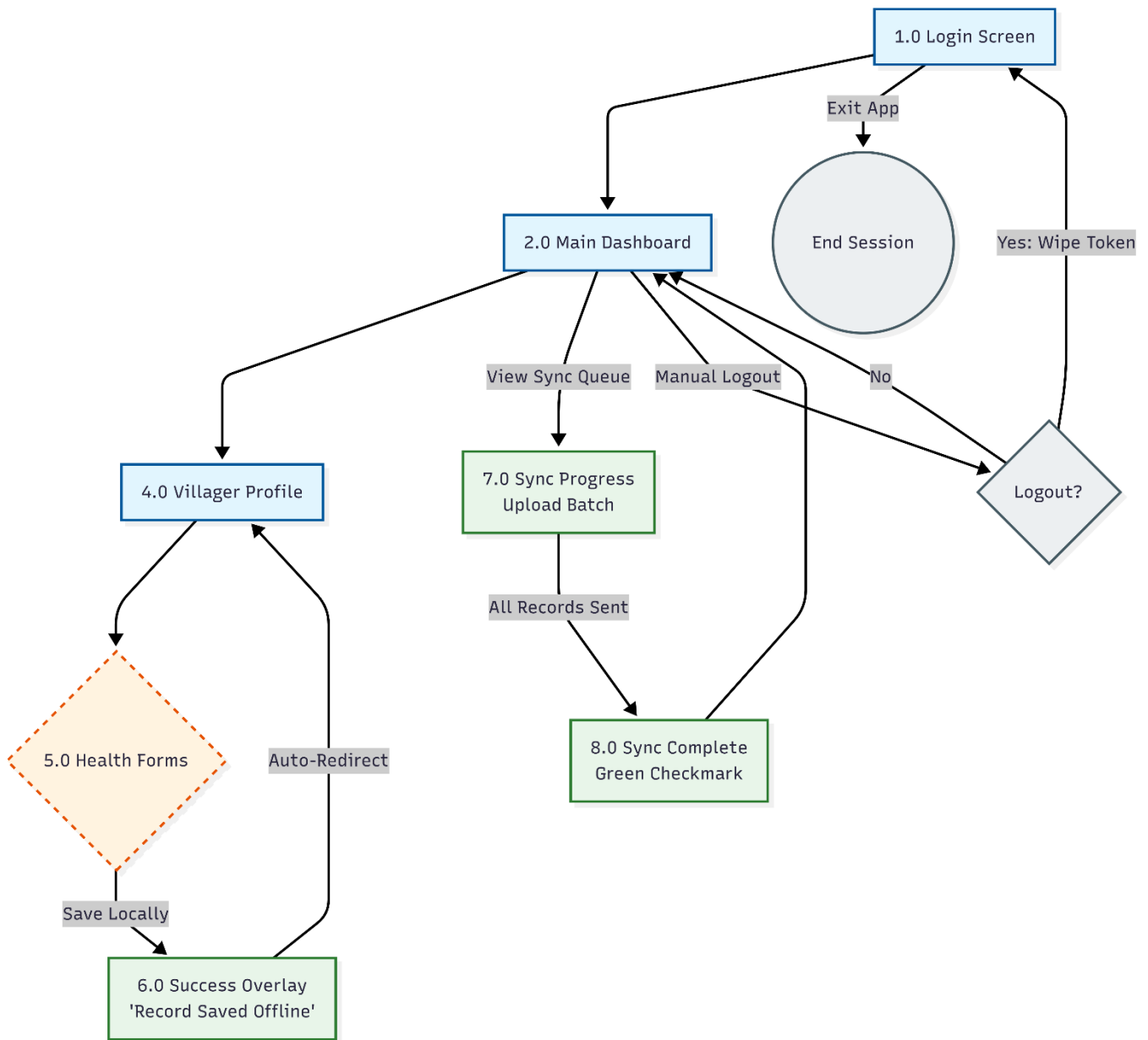


Figure 3: User Interface (UI) Navigation Flow and End-to-End Session Lifecycle

3.5. Sequence Diagram

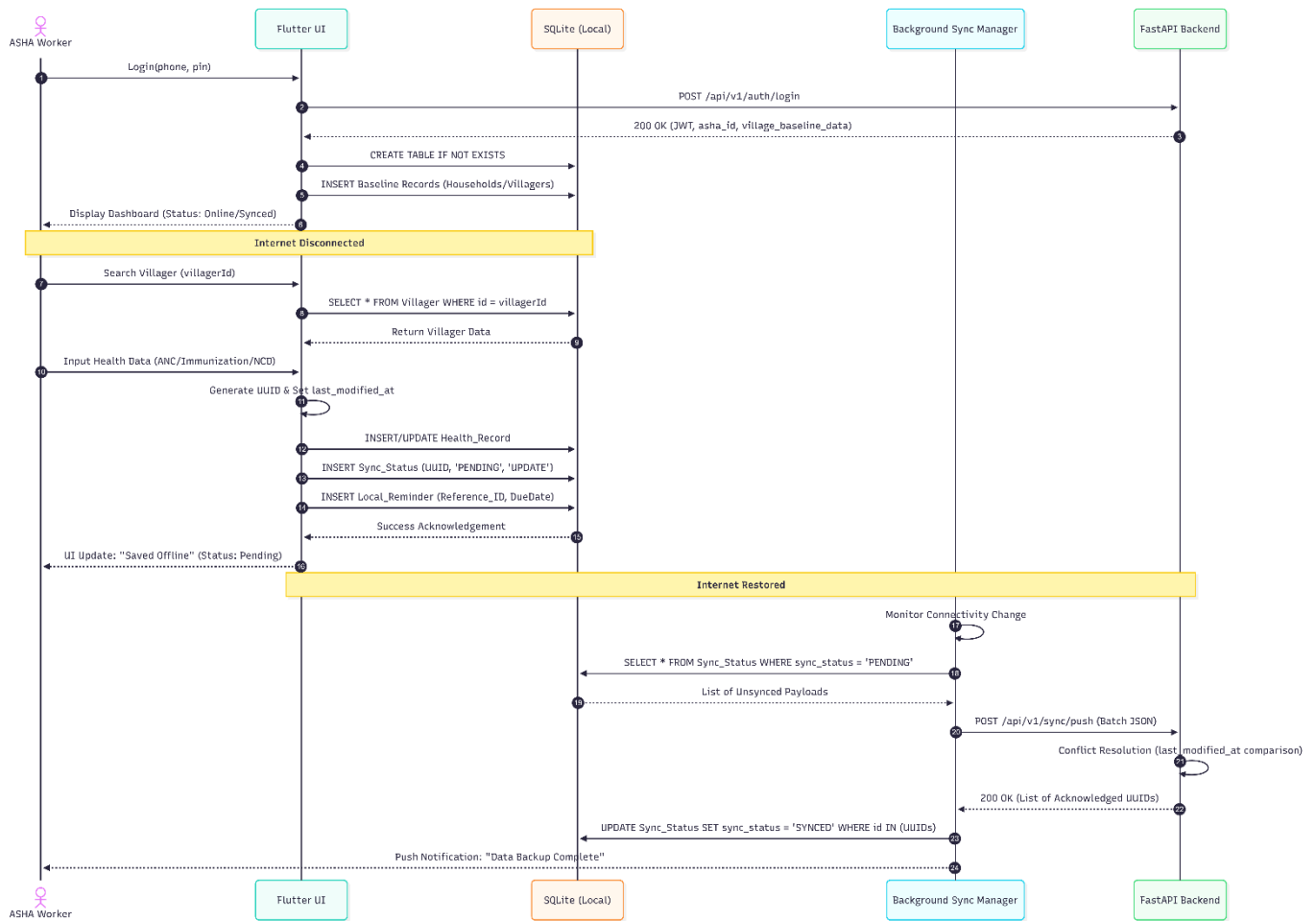


Figure 4: Sequence Diagram of Synchronous and Asynchronous Data Operations during Offline-to-Online Transitions.

4. Implementation Details

This chapter details the technical environment and resource requirements for the ArogyaSathi application.

4.1. Software Specifications

The software stack is designed for a robust, "offline-first" experience using modern, high-speed technologies.

- 4.1.1. Frontend Framework: Flutter 3.10+ — For building the cross-platform mobile user interface.
- 4.1.2. Programming Language: Dart 3.0+ — Used for its efficient handling of background sync tasks.
- 4.1.3. Local Database: SQLite — Provides reliable data storage on the phone when internet is unavailable.
- 4.1.4. Backend API: FastAPI (Python) — A high-speed framework for syncing data to the cloud.
- 4.1.5. Cloud Database: PostgreSQL 15+ — The central repository for all permanent health records.
- 4.1.6. Security: JWT & SSL/TLS — Ensures secure logins and encrypted data transmission.

4.2. Hardware Specifications

These are the minimum requirements to ensure the app runs smoothly on common smartphones used by ASHA workers.

4.2.1. Client-Side (Mobile Device)

- ❖ Processor: 1.8 GHz Quad-core (to handle data processing and UI).
- ❖ RAM: 3 GB Minimum (to support background synchronization).
- ❖ Storage: 500 MB free space (for the local database).
- ❖ OS: Android 8.0 or higher.
- ❖ GPS: Required for household geotagging.

4.2.2. Server-Side (Cloud/Hosting)

- ❖ CPU: Dual-core vCPU (for handling incoming sync requests).
- ❖ RAM: 4 GB.
- ❖ Storage: 20 GB SSD (for fast database operations).
- ❖ Network: 100 Mbps uplink.

4.2.3. System Architecture

The system follows a **Three-Tier Architecture** to separate the user interface from the database logic.

- ❖ Presentation Tier: The Flutter app where the ASHA worker enters data.
- ❖ Logic Tier: The FastAPI server that validates and processes data.
- ❖ Data Tier: The hybrid storage system (Local SQLite + Cloud PostgreSQL).

4.2.4. **Security and Data Integrity**

To protect sensitive health information, the following measures are in place

- ❖ **Offline Persistence:** Data is saved locally the moment a form is submitted to prevent data loss.
- ❖ **Encrypted Sync:** All data sent to the cloud is encrypted via HTTPS to prevent hacking.
- ❖ **Authorized Access:** Only workers with a valid Phone Number and PIN can access the system.
- ❖ **Conflict Resolution:** The system uses timestamps to ensure the most recent update is always saved.

5. Outputs and Reports

This chapter details the primary outputs generated by the **ArogyaSathi** system. The focus is on providing immediate, actionable information to the ASHA worker upon the successful completion of field tasks.

5.1. Real-Time Data Outputs

The system provides immediate visual and data feedback to ensure the user that field records have been successfully captured.

- 5.1.1. Digital Health Profiles: Upon selecting a villager, the system outputs a comprehensive digital summary of their demographics and previous health history directly from the local SQLite database.
- 5.1.2. Service Confirmation Overlays: A critical output triggered after every ANC, Immunization, or NCD form submission. This provides an immediate visual confirmation that the record is "Saved Locally" and assigned a unique tracking ID.
- 5.1.3. Automated Due Lists: The system processes stored birth dates and last visit dates to output a prioritized list of villagers who are currently due for medical services

5.2. Synchronization and Status Reports

These outputs track the data transition from the mobile device to the central server.

- 5.2.1. Sync Status Dashboard: A visual report on the main dashboard showing the count of "Pending" vs. "Synced" records. This acts as the primary audit tool for the ASHA worker.
- 5.2.2. Cloud Acknowledgement Logs: A technical output received from the FastAPI backend. Once a batch is uploaded, the system generates a "Sync Success" notification, confirming that the cloud database (PostgreSQL) has officially accepted the records.

6. CONCLUSION

The ArogyaSathi system demonstrates the feasibility and effectiveness of an offline-first mobile Electronic Health Record (EHR) application designed specifically for ASHA workers operating in rural environments. The project addresses the key challenges faced in rural healthcare data management, particularly unreliable internet connectivity and dependence on paper-based documentation.

By leveraging mobile technology, local data storage, and a synchronization mechanism, the system enables ASHA workers to record patient information, health visits, immunization details, and maternal care data efficiently in the field. The offline-first architecture ensures that essential healthcare data can be captured without continuous network access and synchronized with the backend server when connectivity becomes available.

The implemented modules, including patient registration, health visit tracking, immunization management, reminders, and synchronization services, collectively provide a lightweight yet functional healthcare data management solution. The use of Flutter for the mobile application, SQLite for local storage, and FastAPI for backend services ensures a scalable and maintainable system architecture.

Overall, the ArogyaSathi prototype highlights the potential of mobile health (mHealth) technologies to improve data collection, reporting efficiency, and healthcare service delivery in underserved rural communities.

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- 7.1.3. [Venkat, M. S. The Evolution of Electronic Health Records in India: Progress, Challenges, and Future Prospects \(2020–2025\). International Journal of Health Technology and Innovation, 2025.](#)

7.2. Web Resources

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- 7.2.2. National Health Mission, Government of India: <https://nhm.gov.in>
- 7.2.3. Ministry of Health and Family Welfare – ASHA Programme: <https://nhm.gov.in/index1.php?lang=1&level=1&sublinkid=150&lid=226>

7.3. Tools, Frameworks and Libraries (Referred & Used)

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- 7.3.2. SQLite – Official Documentation: <https://www.sqlite.org/docs.html>
- 7.3.3. FastAPI – Official Documentation: <https://fastapi.tiangolo.com>
- 7.3.4. Dart Programming Language: <https://dart.dev/docs>